



ELECT MAINT- OFF SITE Department
SIMHADRI

Ref:SMTTP/ELECT MAINT-/2022-23/GCB/507143

Date:

11/10/2022(12:10:54)

Subject:Revision of LMI/OGN/OPS/ELEC/036:Guidelines on Generator Circuit Breaker Maintenance and Overhauling

Sub:- Review of LMI/OGN/OPS/ELEC/036 Reg

LMI/OGN/OPS/ELEC/036 on **Guidelines on Generator Circuit Breaker Maintenance and Overhauling**

ISSUE NO: 1, REVISION: 0 **DATE: October 2020** has been reviewed and found in order, No changes are proposed in the existing LMI.

Submitted for kind approval of the competent authority.

Submitted by:

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COS reviewed the corresponding OGN-OPS-ELEC-036 on 02/08/2022 and NO CHANGES ENVISAGED.

Hence LMI periodic review done and no changes proposed.

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कृपया अनुमोदन प्रदान किया जाए।

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On:14/10/2022(10:45:20) **Ref:**SMTTPP/ELECT MAINT-/2022-23/GCB/507143

NTPC Limited

SIMHADRI SUPER THERMAL POWER STATION

LMI/OGN/OPS/ELEC/036
ISSUE NO: 1, REVISION: 0
DATE: October 2022

**Guidelines on Generator Circuit
Breaker Maintenance and
Overhauling**

APPROVED FOR

IMPLEMENTATION BY

HOP (SIMHADRI)

Date: _____

NTPC Limited
SIMHADRI SUPER THERMAL POWER STATION

OPERATION & MAINTENANCE DEPARTMENT

LMI (Name) : Guidelines on Generator circuit breaker Maintenance and overhauling			
LMI No. : LMI/OGN/OPS/ELEC/036			
Issue No.	1	Revision : 0	Date: October 2022

	Name	Designation	Signature
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Reviewed by	R K Jha	GM (O&M)	

Guidelines on Generator Circuit Breaker Maintenance and Overhauling

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1.0 Introduction:

Generator Circuit Breaker (GCB) is specifically designed for high current applications and to protect generator and generator transformer by interrupting full load and short circuit currents. Generator Circuit Breaker shall be suitable for synchronizing the generator to grid. Main functions of the Generator Circuit

Breaker system shall be the following: -

- (a) Startup of the turbo-generator auxiliaries via its auxiliary transformer with the Generator circuit breaker is in open condition
- (b) Carry the full rated current of the generator continuously without exceeding temperature rise as per rating and ensure required insulation level at all time.
- (c) Paralleling of the generator with external network.
- (d) Disconnection of the generator from the system under various loading conditions ranging from no load to full load.
- (e) Interrupts generator-fed and transformer-fed short-circuit currents.

It protects the transformer from generator side faults and generator from system faults by isolating generator from system through opening GCB in few milliseconds and preventing severe damages. In Simhadri GCB exists in unit-1 &2. Arc quenching is by air blast and operation is pneumatic (air). GCB of Simhadri consists of one earth connector (GCB earth switch) and breaker of 3 segregated poles enclosed in conducting chamber ground insulated through shims. Each pole includes main interrupting chamber and auxiliary interrupting chamber. GCB of Simhadri doesn't have GCB isolator (disconnecter). A cooling system which consists of two radiators with overhead fans to cool the compressed air inside the breaker. A compressor plant will provide air with required pressure of 255 bar in high pressure storage tank to 33 bar after pressure reducer into breaker. Being a very critical and expensive piece of equipment, it is necessary that periodic maintenance and overhauling of GCB are carried out following OEM maintenance and overhauling guidelines.

2.0 Superseded documents: Nil

3.0 Safety instructions

All safety and general instructions laid down in the documentation supplied with the equipment must be followed at times by the maintenance personnel. Some of the general safety guidelines applicable to personnel and equipment are as follows:

(a) Personnel:

- Appropriate clothing, gloves, helmet, safety boots, harness etc.
- The personnel concerned must be familiar with the basic working regulations governing a given GCB: mechanical, dielectric, pressure hazards and electrocution etc.

(b) Handling Equipment

- Slings, cloth slings used must have proper valid test certificates and proper marking of tonnage of safe load limit

(c) Tools in General:

- Only use tools appropriate to the type of work to be carried out. OEM supplied tools are available and to be used for GCB for opening of pneumatic system and poles- main and auxiliary chambers.

(d) Working Area:

- Make sure the floor is safe (free from oil, blunt objects, etc.).
- Make sure the confined spaces are properly ventilated before personnel enter.
- Obey the lock out /tag--out and permit to work procedures
- The site must be properly demarcated and kept clear

4.0 REFERENCE DOCUMENTS:

- COS-ISO-00-OGN/OPS/ELEC/036, Rev. No.: 0 dated : September 2020
- O&M Manual of GCB.

5.0 Maintenance plan of Circuit Breaker, Earth Switches

5.1 Maintenance Frequency- Under normal conditions the maintenance plan of circuit breaker and earth switches are as follows

Type of examination	Frequency
Maintenance Visits	Every Week
Inspection/Function and service checks	During unit overhauls
Equipment Overhaul	Every 10 Years or Electrical wear limit by analyzing test reports of breaker

5.2 Activities to be carried out in maintenance of Circuit breaker and its compressor

Activities	Maintenance Frequency		
	Maintenance Visit (weekly)	Inspection (unit overhaul)	Overhaul (10years)
1.Operation Counter Reading	Yes	Yes	Yes
2. Overall inspection: appearance (corrosion, paintwork, signs of overheating).		Yes	Yes
3. Inspect the insulating envelopes of poles.			Yes
4. Check Air pressure level	Yes	Yes	Yes
5. Check Air densimeter thresholds. Then adjust pressure to its rated value.			Yes
6. Taking Timing of contacts		Yes	Yes
7. Measure resistance of main contacts.		Yes	Yes
8. Check tightening of parts not subjected to pressure (frame - deck - cubicle).			Yes
9. Check tightening of low voltage terminals		Yes	Yes
10. Check all relays and latch operations.		Yes	Yes

11. Check doors seals, side panel seal and roofs seals.		Yes	Yes
12. Check the complete control circuit. Replace components if required.		Yes	Yes
13. Check the CB connector rod assy. Change if required			Yes
14. Check the condition of the motor—driven fans		Yes	Yes
15. Check the tightness of CB envelope			Yes
16. Check for any abnormal sound in cooling fan motor.(if any replace the motor)	Yes	Yes	Yes
17. Check insulation resistance of all power and control circuits		Yes	Yes
18. Checking of all interlocks and alarms by actual operations to the extent possible		Yes	Yes
19. Visual Inspection of main & aux chamber contacts, blast valves, Insertion resistor, camber insulating enclosure, muffler filter etc.			Yes
20. Main & Aux chamber replacement of sliding contacts of moving contact, pressurized insulators, cylinders and piston (of compressor).			Yes (if required)
21. Tightness checking, replacement of suction filters, Inspection of electro valves, pneumatic control and safety valves		Yes	Yes
22. HP receiver inspection, maintenance of reducing panel, maintenance of automatic HP drain, oil change, Cleaning of valves , Cleaning of bleeding pots, Inspection of automatic bleeding valves.		Yes	Yes
23. Check operation of locking key and electrical interlocking.		Yes	Yes

6.0 Check list format for maintenance weekly, unit overhaul and GCB overhaul.

6.1 Check list format for Maintenance weekly.

Description	Reading
1. No of Operations of breaker	
2. Air pressure : Inside HP receiver Tank	
3. Air Pressure: After Pressure reducer	
4. No of openings of pressure reducer valve	
5. Compressor -1 No of operations	
6. Compressor-1 Running Hours	
7. Compressor-2 No of Operations	
8. Compressor-2 Running Hours	
9. Spare Pole air pressure	
10. Check for any abnormal sound in cooling fan motor.(if any replace the motor)	

6.2 Check list format for Maintenance during unit overhauling.

Description	Reading/remark
1. Timing of operation of circuit breaker.	
2. Overall inspection: appearance (corrosion, paintwork, signs of overheating).	
3. Contact Resistance measurement.	
4. Check tightening of low voltage terminals	
5. Check all relays and latch operations.	
6. Check doors seals, side panel seal and roofs seals.	
7. Check the complete control circuit. Replace components if required.	
8. Check the condition of the motor—driven fans	
9. Check insulation resistance of all power and control circuits	
10. Checking of all interlocks and alarms by actual operations.	
11. Tightness checking, replacement of suction filters, Inspection of electro valves, pneumatic control and safety valves	
12. . HP receiver inspection, maintenance of reducing panel, maintenance of automatic HP drain, oil change, Cleaning of valves , Cleaning of bleeding pots, Inspection of automatic bleeding valves.	
13. Check operation of locking key and electrical interlocking.	

6.3 Check list format for Maintenance during GCB Overhaul.

Description	Reading/remark
1. Timing of contacts	
2. Overall inspection: appearance (corrosion, paintwork, signs of overheating).	
3. Contact Resistance measurement.	
4. Check tightening of low voltage terminals	
5. Check all relays and latch operations.	
6. Check doors seals, side panel seal and roofs seals.	
7. Check the complete control circuit. Replace components if required.	
8. Check the condition of the motor—driven fans	
9. Inspect the insulating envelopes of poles.	
10. Checking of all interlocks and alarms by actual operations.	
11.Tightness checking, replacement of suction filters, Inspection of electro valves, pneumatic control and safety valves	
12. . HP receiver inspection, maintenance of reducing panel, maintenance of automatic HP drain, oil change, Cleaning of valves , Cleaning of bleeding pots, Inspection of automatic bleeding valves.	
13. Check operation of locking key and electrical interlocking.	
14. Check the CB connector rod assy. Change if required	
15.Visual Inspection of main & aux chamber contacts, blast valves, Insertion resistor, camber insulating enclosure, muffler filter etc.	
16. Main & Aux chamber replacement of sliding contacts of moving contact, pressurized insulators, cylinders and piston (of compressor).	

7.0 RESPONSIBILITY CENTRE FOR IMPLEMENTATION:

Switchyard area maintenance engineer for implementation and overall responsibility lies with sectional head of electrical maintenance.

8.0 EXCEPTION REPORTING AND DISPENSATION OF DEVIATIONS

Any exceptions / deviations shall be reported to HOD (EM).

9.0 RECORD KEEPING:

Records will be maintained by Switchyard area maintenance engineer.

10.0 REVIEW:

This LMI document will be reviewed once in every two years by Head of the Project, NTPC Simhadri.